Multiplying and Dividing in Scientific Notation

Question Bank 4.02

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(2\times10^{3})(4\times10^{5})$

2. $\frac{8.8\times10^{9}}{2.2\times10^{4}}$

3. $(6\times10^{4})(5\times10^{7})$

4. $\frac{1.5\times10^{6}}{5\times10^{9}}$

5. $(3.5\times10^{-6})(4\times10^{-3})$

6. $\frac{4.8\times10^{-2}}{1.6\times10^{-7}}$

Real quantities.

7. Three meteorites have masses 1.1×10^{2} , 6.8×10^{2} and 8.4×10^{-2} grams.

a. How many times as heavy as the lightest one is the heaviest one? Round to the nearest whole number.

b. How many times as heavy as the lightest one is the 1.1×10^{2} gram one? Round to the nearest whole number.

8. A data center processes about 4.5×10^{6} records per hour.

a. How many records does it process in 2.4×10^{2} hours?

b. A second center processed 9.0×10^{9} records in 1.8×10^{3} hours. How many records per hour is that?

Working backwards.

9. Find the missing number in each equation.

a. $(3\times 10^{n})(2\times 10^{4})=6\times 10^{11}$

b. $\frac{9.6\times 10^{8}}{n\times 10^{3}}=1.2\times 10^{5}$

c. $(5\times 10^{4})(4\times 10^{6})=$ _____

Reading someone else's work.

10. Tyree multiplies $(2.8\times 10^{3})(9.7\times 10^{6})$ and writes 27.16×10^{9}. Is his value correct? Is his answer written in scientific notation? Write the finished answer.

11. Which is the value of $\frac{7.5\times10^{-4}}{2.5\times10^{2}}$?

- ○ A. 5×10^{-6}
- ○ B. 3×10^{-6}
- ○ C. 3×10^{6}
- ○ D. 3×10^{-2}

12. Select ALL of the expressions whose value is 6×10^{8} .

- ☐ A. $(1.5\times10^{4})(4\times10^{4})$
- ☐ B. $\frac{3\times10^{10}}{5\times10^{1}}$
- ☐ C. $(2\times10^{3})(3\times10^{5})$
- ☐ D. $\frac{1.2\times10^{12}}{2\times10^{4}}$
- ☐ E. $(2\times10^{3})(3\times10^{15})$
- ☐ F. $\frac{1.2\times10^{12}}{2\times10^{3}}$